Analysis of the Relationship between Body Mass Index (BMI) and Fasting Blood Glucose

Application of Simple Linear Regression Model on Adult Synthetic Dataset

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Preventing Disease, Prolonging Life, and Promoting Health Through the Organized Efforts of Society

Background: Metabolic Risk & Anthropometry

Urgency of Diabetes Mellitus

Non-communicable disease with serious complications. An essential early monitoring indicator is Fasting Blood Glucose (FBG).

Role of Indicator (BMI)

Body Mass Index is a cheap, simple, and valid anthropometric indicator to map nutritional status and metabolic risk in adults.

Rationale of Analysis

Epidemiologically, increased BMI correlates with insulin resistance. Quantitative analysis is needed to model the magnitude of FBG change for each unit increase in BMI.

Problem Formulation & Purpose

Problem Formulation:

Is there a significant relationship between Body Mass Index (BMI) and Fasting Blood Glucose in the adult synthetic dataset?

Method Application

Implementing the R programming language for simple linear regression reproducibly.

Correlation Assessment

Assessing the direction and strength of the statistical relationship between BMI and Fasting Blood Glucose.

Output Interpretation

Analyzing the intercept, slope, p-value, R-squared, and conducting residual assumption checks.

Methodology and Data Approach

Stage 1: Design & Data Source

Realistic synthetic dataset created via R programming script (N = 250 observations). Internally compiled so that the analysis is 100% reproducible without external files.

Stage 2: Data Characteristics

Includes variations in BMI, physical activity, and DM history. Ethical Note: Data is purely for computational statistics learning, not for causal diagnosis.

Stage 3: Software

Primary statistical analysis is executed using the lm() function within the R Core Team ecosystem.

Based on Slide C

Research Variable Mapping

Independent Variable (X)	Dependent Variable (Y)
Body Mass Index (BMI)	**Fasting Plasma Glucose (FPG)**
Data Type: Continuous Numeric	**Data Type:** Continuous Numeric
Unit: kg/m^2	**Unit:** mg/dL

Method Suitability: Simple Linear Regression was chosen because both variables are continuous numeric, designed specifically to assess the impact of one independent variable on one dependent variable.

Simple Linear Regression Model Architecture

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Fasting Blood Glucose
(Dependent Variable)

Body Mass Index
(Independent Variable)

Error / Residual
(Variation in Y not explained by the model)

Intercept
(Intersection point; predicted Y value when X = 0)

Slope / Regression Coefficient
(Rate of change in Y for every 1 unit of X)

Summary of Data Characteristics (N = 250)

Average BMI

27.10

± 4.16 kg/m² (Range: 18.2 - 38.8)

Average Fasting Blood Glucose

112.58

± 14.63 mg/dL (Range: 78.4 - 150.6)

BMI Group

Overweight: 46.0%

Obesity: 26.0%

Normal: 25.2%

Underweight: 2.8%

GDP Category

Prediabetes: 62.8%

Normal: 20.0%

Diabetes: 17.2%

Regression Estimation Results

$$Y = 59.992 + 1.941X$$

Key Parameters & Interpretation

Slope BMI (β_1) **1.941**	Each 1 kg/m² increase in BMI increases average FBG by 1.941 mg/dL.
P-value **< 0.001**	Statistically significant positive relationship at $\alpha = 0.05$.
R-squared (R^2) **0.305**	The BMI model is able to explain 30.5% of the variation in Fasting Blood Glucose.
F-statistic **108.684**	The overall model is statistically feasible.

Model Diagnostics & Residual Examination

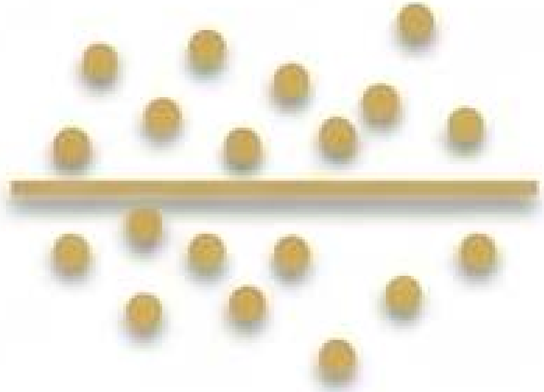
Residual Histogram

Distribution shape is relatively balanced and symmetric around zero.

Q-Q Plot

Residuals follow the normal diagonal line without extreme deviations.

Residual vs Fitted



Variance is randomly distributed, without curved patterns (homoscedasticity is met).

Formal Normality Test

Shapiro-Wilk Test: p-value = 0.763 ($p > 0.05$)

Conclusion: Residual distribution does not significantly deviate; model satisfies the assumption of simple linear normality.

Discussion of Results & Model Limitations

Substantial Interpretation

A positive and linear relationship between BMI and FPG is statistically proven.

The model's explanatory power (R^2 30.5%) is considered logical for public health data, given that glucose levels are multifactorial.

Intercept Note: The value 59.992 is purely a mathematical component of the regression line. Biologically, BMI = 0 is unrealistic (not a clinically observed value).

Research Limitations

This simple regression model does not control for confounding variables (age, physical activity, genetic history, gender).

The dataset is synthetic (for the purpose of R programming final exam). The output is not intended for drawing pure causality or as a basis for clinical diagnosis.

Conclusion

There is a positive and statistically significant relationship between Body Mass Index (BMI) and Fasting Blood Glucose (FBG).

$$Y = 59.992 + 1.941X$$

The prediction model shows good basic validity, evidenced by a significance p-value < 0.001 and fulfillment of the residual validity test (Shapiro-Wilk $p = 0.763$).

Methodologically, this approach successfully demonstrated the implementation of computational statistics using R programming accurately and reproducibly.

THANK YOU

Discussion and Q&A Session

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